Kota Yoshimura

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Education

- **University of Notre Dame** — Notre Dame, IN, US
  Phd in Physics — *Aug 2021 - Present*
- **Shizuoka University** — Shizuoka, Japan
  Bachelor of Science in Physics — *April 2017 - March 2021*

Publication

- **Defect-nucleated charge density wave in monolayer** $NbSe_2$:
  F. Cheng et al., "Defect-nucleated charge density wave in monolayer 1H-NbSe," Phys. Rev. B 114, 245424 (2026).
  https://journals.aps.org/prb/abstract/10.1103/d1k3-m1sl
- **RHEED pattern classification by a convolutional neural network for the growth of chalcogenide thin films and nanostructures** :
  Muetzel, Nathan, et al. "RHEED pattern classification by a convolutional neural network for the growth of chalcogenide thin films and nanostructures." AIP Advances 16.3 (2026).
  https://pubs.aip.org/aip/adv/article/16/3/035317/3382644
- **g-Factor Enhanced Upper Critical Field in Superconducting PdTe2 due to Quantum Confinement** :
  Yoshimura, Kota, et al. "g-Factor Enhanced Upper Critical Field in Superconducting PdTe2 due to Quantum Confinement." arXiv preprint arXiv:2508.07547 (2025).
  https://arxiv.org/abs/2508.07547
- **Probing berry curvature in magnetic topological insulators through resonant infrared magnetic circular dichroism** :
  Bac, Seul-Ki, et al. "Probing berry curvature in magnetic topological insulators through resonant infrared magnetic circular dichroism." Physical Review Letters 134.1 (2025): 016601.
  https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.134.016601
- **Molecular beam epitaxy growth of axion insulator candidate** $EuIn_2As_2$:
  Karim, Muhsin Abdul, et al. "Molecular beam epitaxy growth of axion insulator candidate $EuIn_2As_2$." Physical Review Materials 7.10 (2023): 104202.
  https://journals.aps.org/prmaterials/abstract/10.1103/PhysRevMaterials.7.104202
- **Bootstrap method in harmonic oscillator**:
  Aikawa, Yu, Takeshi Morita, and Kota Yoshimura. "Bootstrap method in harmonic oscillator." Physics Letters B 833 (2022): 137305.
  https://www.sciencedirect.com/science/article/pii/S0370269322004397/
- **Application of bootstrap to a theta term**:
  Aikawa, Yu, Takeshi Morita, and Kota Yoshimura. "Application of bootstrap to a term." Physical Review D 105.8 (2022): 085017.
  https://journals.aps.org/prd/abstract/10.1103/PhysRevD.105.085017
- **Propagation of a peak-truncated Gaussian pulse in an inverted coupled-resonator-induced transparency system**:
  Sugio, Daiki, et al. "Propagation of a peak-truncated Gaussian pulse in an inverted coupled-resonator-induced transparency system." Optics Communications (2022): 128466.
  https://www.sciencedirect.com/science/article/pii/S0030401822003224
- **Development of the pulse peak from peak-truncated Gaussian optical pulses in a serial array of high-Q ring resonators**:
  Tomita, Makoto, et al. "Development of the pulse peak from peak-truncated gaussian optical pulses in a serial array of high-q ring resonators." Physical Review A 102.4 (2020): 043507. APA
  https://journals.aps.org/pra/abstract/10.1103/PhysRevA.102.043507

Presentation

- **Violation of Pauli limit in chemically tuned non-centrosymmetric superconducting (Sn,Ge,In)Te thin films**
  Quantum Fluids and Solids (QFS), Poster Presentation
- **g-Factor-Enhanced Upper Critical Field in Superconducting** $PdTe_2$ **Due to Quantum Confinement**
  APS March meeting, Session M34: Low Dimensional Superconductivity I, Mar 18 2026
- **Non-Monotonic Behavior of the Critical Field of Superconducting PdTe2 thin films**
  APS March meeting, Session Q24: Two-dimensional Superconductors, van der Waals Interactions, and the Dichalcogenides I, Mar 2
- **Achieving and tuning an unconventional superconducting state in PdTe2 grown by CVD**APS March meeting
  Session G15: Superconducivity: Thermodynamic and Transport Properties II — *Mar 5 2024*
- **Poster Presentation, Fundamentals of Quantum Materials Winter School** — University of Maryland
  Kota Yoshimura, Infrared Spectroscopy of MBE grown MnBi2Te4 at high magnetic field. — *Jan 16 2023*

- **Investigating the Quantum system using Bootstrap Method** — The Physical Society of Japan
  Yu Aikawa, Takeshi Morita, Kota Yoshimura. 2021 (76th) Annual Meeting, Oral Presentation — *March 15 2021*
- **BB84 Protocol (Quantum Information)** — Shizuoka University
  Kota Yoshimura, Ryoto Hori. Poster Presentation — *February 12 2020*

Digital Skills

- **Languages**: Mathematica, Fortran 90-95, Python, Julia, MatLab, C/C++.

Teaching Experiences

- **Teaching Assistant** — University of Notre Dame, IN, US
  TA for Quantum Mechanics — *Jan 2025 – May 2025*
- **Teaching Assistant** — University of Notre Dame, IN, US
  TA for Condensed Matter Physics, Quantum Mechanics — *August 2021 – May 2023*
- **Teaching Undergrad Students** — University of Notre Dame, IN, US
  Private Tutor for two semesters, taught Physics — *October 2021 – May 2022*
- **Teaching High School Students** — Shizuoka, Japan
  Private Tutor for two years, taught math — *December 2017 – August 2019*

Workshop/conference

- **Quantum Fluids and Solids (QFS)** — University of Notre Dame, South Bend, IN
  QFS2026 — *Aug 2–8, 2026*
- **APS March Meeting 2026** — Convention Center, Denver, CO
  American Physical Society annual meeting — *March 16–20, 2026*
- **APS March Meeting 2025** — Annaheim Convention Center, CA,US
  American Physical Society annual meeting — *March 16 - March 21, 2025*
- **Controlling Low-Dimensional Quantum Systems** — University of Notre Dame, IN, US
  Condensed Matter Workshop — *October 19 - October 20 , 2024*
- **APS March Meeting 2024** — Minnealpolis Convention Center, MN,US
  American Physical Society annual meeting — *March 3 - March 8, 2024*
- **Fundamentals of Quantum Mechanics Winterschool** — Univeristy of Maryland, MD,US
  Focus on 2D material — *Jan 16 -Jan 20, 2023*
- **PIKIMO 12** — University of Notre Dame, IN, US
  High Energy Theory, Phenomenology Conference — *April 30, 2022*
- **Hamamatsu Photonics** — Shizuoka, Japan
  Workshop on photonics research — *July 29, 2017*

Internship

- **National Institute for Materials Science (NIMS)** — Tsukuba, Japan
  Learned CVD method at 2D Quantum Materials Group and collaborated. — *June 2023 – December 2023*
- **Keiei-Sanbou, Zoho Japan** — Shizuoka, Japan
  Internship at an IT company -Created multiple interactive websites -Learned Marketing strategies.August 2019 – December 2019

Experimental Skills

- Electron Beam Lithography, Photo Lithography, Atomic Force Microscope, Raman Spectroscopy, Metal Deposition, Scanning Electron Microscopy, XRD, Magnetotranport at low temperatures and high magnetic field, CVD.